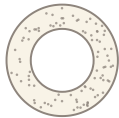


a Cine lacks a material label

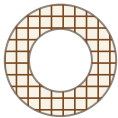
Routine cine



boundaries +
weak texture



Tagging



prepared grid
(material pattern)



DENSE



phase-encoded
displacement



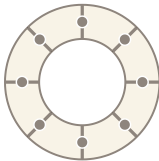
material label carried by the signal

Cine supplies boundaries and weak texture only;
correspondence inside the wall must be inferred.

b Same masks, two material maps

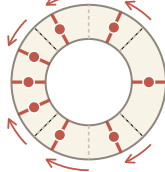
Mapping 1

$$\theta \mapsto \theta$$



Mapping 2

$$\theta \mapsto \theta + a \sin \theta$$



identical
masks

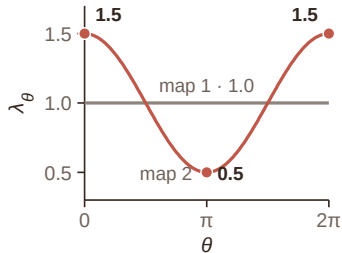
$$\text{Dice} = 1 \cdot \text{HD} = 0 \cdot r \mapsto r \cdot a = 0.5$$

— reference position

● mapped point

Same eight material points in both annuli.

c Stretch differs by $\pm 50\%$



$$\lambda_\theta = 1 + a \cos \theta$$

range, map 2 **0.5–1.5**

mean over θ , both maps **1.0**

orientation preserved for **$|a| < 1$**